MODEL & DATA CARD

EVENT: SCIBLITZ AI HACKATHON 2026

PROJECT: BIOFOLD | TEAM: CTRL+ALT+DEFEAT

1. PRE-TRAINED MODELS UTILIZED

The BioFold pipeline operates via a zero-shot, multi-agent inference architecture, leveraging two distinct foundational models:

- **Llama-3-8B-Instruct:** Provider: Meta. Synthesizes chain-of-thought clinical rationales and generates the primary amino acid sequence (60-100 residues) under strict thermodynamic constraints via a Dynamic Context Injector.
- **ESMFold (esm2_t36):** Provider: Meta. An evolutionary-scale transformer that performs high-confidence prediction of 3D atomic coordinates (PDB generation) directly from the sequence.

2. DATASETS & GROUND TRUTH

BioFold operates on foundational training corpuses to ensure generalized biological reasoning without requiring custom fine-tuning datasets:

- **Llama-3 Corpus:** Utilized for broad biochemical reasoning and natural language processing.
- **UniRef50 (UR50D):** The evolutionary sequence database utilized by ESMFold to map protein language and predict folding pathways.
- **Context Injection:** Target toxin atomic structures (e.g., Fentanyl, Digoxin) are hardcoded into the backend via standard pharmacological databases (PubChem) to enforce absolute chemical ground truth.

3. KNOWN LIMITATIONS

- **Static Prediction:** ESMFold predicts a static, idealized 3D topology. It does not currently simulate dynamic folding pathways or complex multi-body interactions in blood plasma.
- **Affinity Estimation:** Absolute binding affinity (Kd) cannot be mathematically guaranteed without molecular dynamics (MD) simulations or in vitro wet-lab validation.
- **Context Limits:** The pipeline is highly optimized for single-domain scavenger proteins. Generating massive complexes exceeds the current generative prompt constraints.

4. ETHICAL CONCERNS & BIOSECURITY

- **Dual-Use Dilemma:** De novo protein engineering inherently possesses dual-use characteristics. While designed for therapeutic scavengers, the mechanics could theoretically be misused.
- **Mitigation Strategy:** The system utilizes a constrained prompt restricted specifically to pharmacological sequestration and hydrophilic plasma solubility.
- **Clinical Disclaimer:** All structural output is strictly for in silico research purposes. Synthesized sequences must undergo rigorous in vitro screening and FDA-standard trials before practical application.